CONNEXUS | 2026
black sea tech forum

Navigating the Future with Autonomous Ships

Global Programs · COLREGs · Model-Based Path Planning · Sailing Drones · Embedded Validation



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ICT CLUSTER VARNA

BSMEPA
BULGARIAN SMALL AND MEDIUM ENTERPRISES PROMOTION AGENCY

BAPHA

BRAIN++
AI FACTORY

ARC FUND

Short Profile



Education

- B.Sc. at Istanbul Technical University (ITU) in Electronics Engineering
- B.Sc. at ITU Control Engineering
- M.Sc. at ITU Electrical Engineering
- PhD at University of Waterloo (UW), Mechanical & Mechatronics Engineering, Canada

Thesis (GM Funded): Developing Collaborative Driving Strategies

Professional experience

- Principal Researcher at Samsung Korea 2023- March 2026
- 3 years with FEV GmbH as project engineer in software development and testing
- 15 months with SAFRAN Energy Efficient Aircrafts
- 10 years as a Professor at Istanbul Technical University
- 4 years with UW Mechanical and Mechatronics Engineering

Position:

Associate Professor Dr. Ismail Bayezit

Current Responsibilities

- Associate Professor, Istanbul Technical University (ITU), Faculty of Aeronautics and Astronautics
- Deputy Director, ITU–Turkish Aerospace Industries Center of Excellence for Aircraft Research
- Director, ITU Model-Based Design and Control Laboratory

Areas of Expertise

- Model based control & testing of autonomous vehicles
- Autonomous Berthing/Unberthing of Ships
- Real-time validation of COLREGS Scenarios
- Model based SW engineering, Modeling, Simulation, Auto Code Generation (Matlab/Simulink/AMESim)
- HIL Systems & Real-Time Validation (dSPACE, SpeedGoat, NI cRIO, FAAR ECU, Ecotrons)
- Embedded Systems, Software Quality, CMMI
- Develop Hybrid and Electrical Vehicle Technologies

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Why Maritime Autonomy — and Why Now

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~90% of World Trade by Sea

≈11 billion tonnes/year moved by ~105,000 merchant ships and 1.9 M seafarers (UNCTAD / ICS)

75–96% of Incidents: Human Factor

Fatigue, inattention and misjudged COLREGs encounters dominate casualty reports (EMSA, Allianz)

IMO MASS Code — 2025/26

Non-mandatory code adopted at MSC 110 (2025); mandatory goal-based code targeted for 2032

Crew Shortage: 90,000 officers by 2026

BIMCO/ICS Seafarer Workforce Report — autonomy and remote ops relieve a structural gap

IMO Degrees of Autonomy

Degree 1

Automated processes & decision support — crew on board

Degree 2

Remotely controlled — seafarers on board

Degree 3

Remotely controlled — no seafarers on board

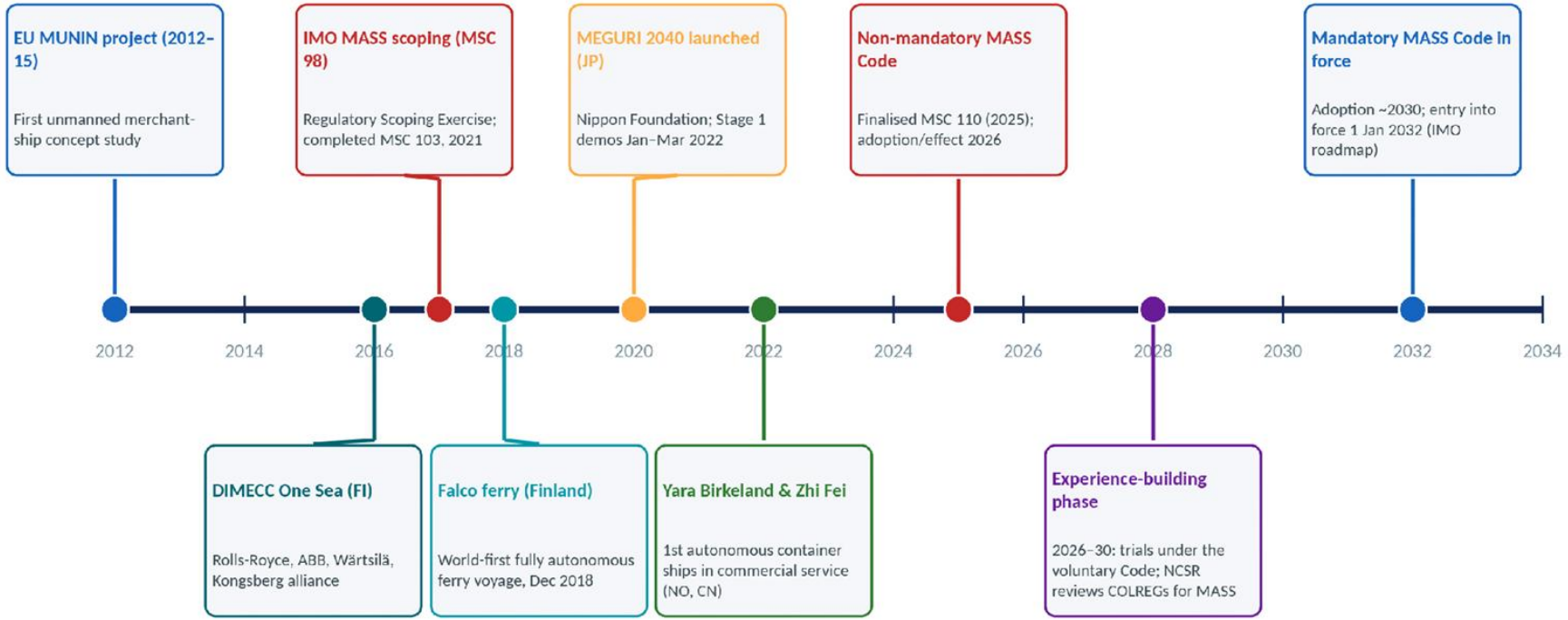
Degree 4

Fully autonomous — system decides & acts alone

Source: IMO MSC.1/Circ.1638 Regulatory Scoping Exercise

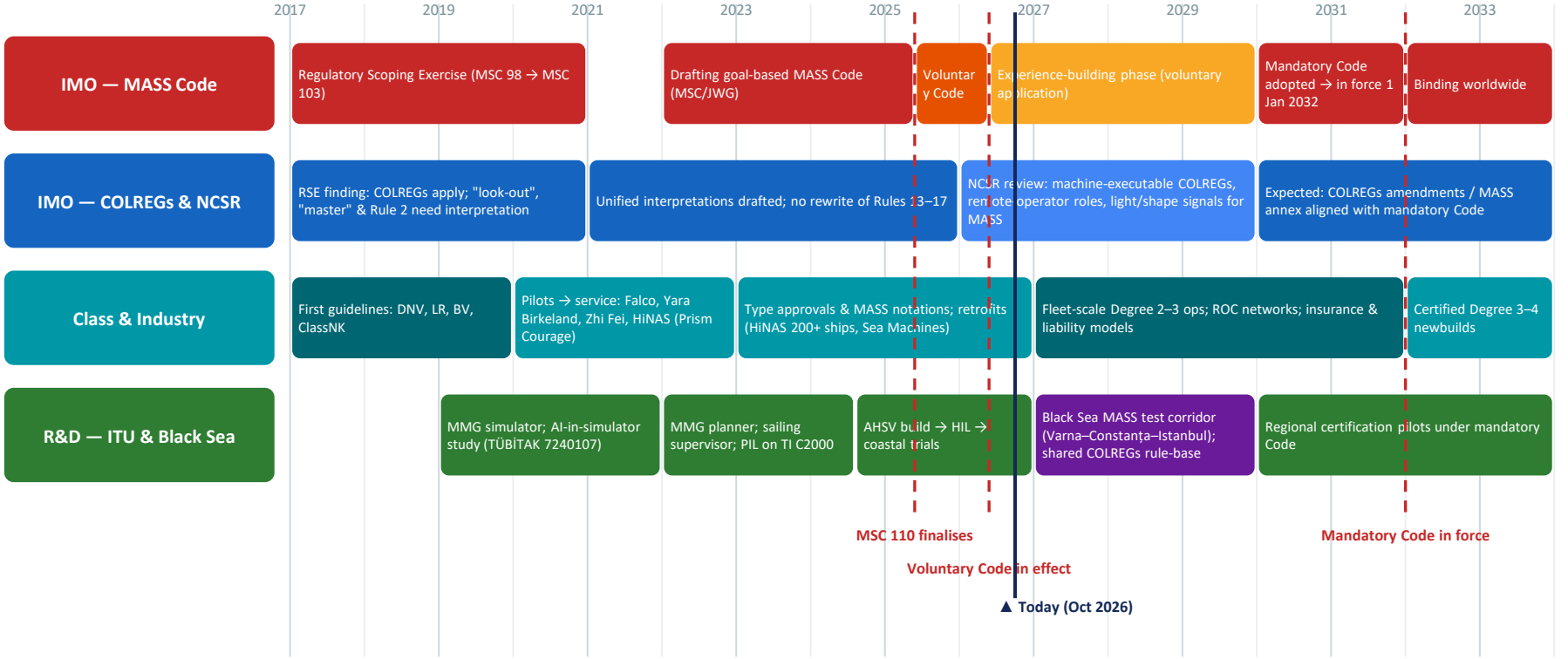
Autonomy is a spectrum — most 2026 deployments operate at Degree 1–2 with Degree 3 trials in Norway, Japan and Korea.

Global Milestones: Two Decades of Autonomous Shipping



Trend: Nordic pilots (2016–22) → Asian scale-up (2020–25) → IMO regulation (voluntary 2026 → mandatory 2032). The Black Sea can enter at the regulation-ready stage.

IMO Roadmap: MASS Code, COLREGs & Readiness



Expectation: COLREGs will not be rewritten for robots — the MASS Code requires compliance and NCSR supplies interpretations. Hence the certifiable asset is a machine-executable, model-aware COLREGs engine — exactly where ITU R&D sits.

Europe: From Pilots to Commercial Operation

🇳🇴 Norway — Yara Birkeland

Kongsberg · Yara · Massterly (2022–)

- 120 TEU, 80 m, 7 MWh battery, zero-emission
- Herøya → Brevik → Larvik fertiliser route
- Removes ~40,000 diesel truck trips / year
- Remote Operations Centre in Horten
- Crew reduced stepwise toward Degree 3
- Sister projects: ASKO sea-drones (Oslofjord)

🇫🇮 Finland — Falco & One Sea

Finferries · ABB · Rolls-Royce (2018–)

- Falco (Rolls-Royce + Finferries): 1st fully autonomous ferry voyage, Turku 2018
- Suomenlinna II (ABB Ability Marine Pilot): remote-piloted Helsinki ferry 2018
- One Sea alliance: ABB, Wärtsilä, Cargotec, Kongsberg
- Jaakonmeri test area — open autonomous test site
- Spin-off: Finnish ISTLAB & Åboat trials
- Target: autonomous commercial fleet by 2025

🇪🇺 EU — AUTOSHIP & Successors

Horizon 2020 → Horizon Europe (2019–)

- AUTOSHIP: €20 M, 2 demos — Eidsvaag Pioner (NO short-sea), Zulu 4 (BE inland)
- Full autonomy stack + remote centre + cyber
- EMSA MASS risk & regulatory studies
- Horizon Europe: SEAMLESS, AEGIS, MOSES
- Belgium/Netherlands inland autonomy corridors
- Lesson: regulation co-designed with pilots

European model: public-private demonstration vessels + national test areas + early IMO engagement — directly transferable to Black Sea ports.

Japan: MEGURI 2040 — The World's Largest Programme

MEGURI 2040

The Nippon Foundation · 50+ companies (NYK, MOL, Mitsui E&S, Furuno, JRC, Yanmar, Japan Radio)

5 ships
Stage 1 full-scale demos (2022)

50%
of domestic fleet autonomous by 2040

¥1 tn
annual economic impact target

Stage 1 2020–22	5 full-scale demonstrations across container, ferry, RoPax and small passenger vessels — world-first set of fully autonomous coastal voyages incl. berthing
Stage 2 2022–25	Commercial-grade systems on 4 vessel types (container, RoPax, tug/ferry); 1 Remote Ops Centre; ClassNK guidelines & ISO/IEC standardisation
Stage 3 2026–40	Commercialisation: 50% of coastal fleet with MASS systems; export of standards (ClassNK, JSTRA)

Stage 1 vessel	Type	Route / demonstration (2022)
Mikage (MOL / Mitsui E&S)	194-TEU container	Tsuruga → Sakai 270 km; auto-berthing with drone mooring line
Soleil (Mitsui OSK Ferry)	Car ferry 222 m	Shinmoji → Iyonada 240 km, 26 kn; fully autonomous incl. berthing
Sunflower Shiretoko (MOL)	RoPax 190 m	Tomakomai → Oarai 750 km, 18 h autonomous navigation
Suzaku (NYK / JMU)	749-GT container	Tokyo Bay → Tsumatsusaka 790 km; remote support from Tokyo ROC
Sea Friend Zero	Small passenger	Sarushima, Yokosuka — AI collision avoidance in dense traffic

Technology stack demonstrated

- Sensor fusion: radar + AIS + LiDAR + IR/visible cameras
- AI collision-avoidance with COLREGs rule engine
- Auto-berthing using RTK-GNSS & lidar docking
- Satellite + 5G remote-monitoring centre (Tokyo)
- Cyber-security & redundancy per ClassNK MASS guideline
- Human-machine interface for Degree 2 → 3 transition

Japan's lesson: a national foundation convening shipowners, yards, electronics firms and the classification society — not a single tech vendor.

Asia-Pacific: Korea, China & Singapore Scale Up

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🇰🇷 South Korea — KASS

MOF · MOTIE · KRISO · HD Hyundai · Samsung HI (2020–25)

- KASS: ₩160 bn (~$120 M) national programme
- Target: IMO Degree 3 for coastal, Degree 2 ocean
- HD Hyundai Avikus HiNAS 2.0: 1st transoceanic autonomous LNG carrier voyage (Prism Courage, 2022)
- Samsung SAS, Hanwha Ocean "SAMSUNG Autonomous Ship"
- Ulsan test-bed & remote control centre

🇨🇳 China — National MASS Drive

CSSC · COSCO · Yunzhou · Navigation Brilliance

- "Zhi Fei" (2022): 300 TEU autonomous feeder, Qingdao–Dongjiakou commercial service
- Navigation Brilliance “Zhi Teng” 1.0/2.0 autonomy systems
- Yunzhou (Zhuhai) 1,000+ USV units; Wanshan test-field
- MSA MASS rules & Qingdao/Zhuhai test zones
- State-led: 14th Five-Year Plan smart shipping

🇸🇬 Singapore — Smart Port MASS

MPA · ST Engineering · PSA (2019–)

- MPA MASS Roadmap + Maritime Innovation Lab
- ST Engineering autonomous harbour craft trials
- Tuas Mega-port: automated quay & vessel coordination
- Keppel / M1 5G remote-ops pilot tugs
- Regulation: MASS technical reference TR 2023

Dimension	Korea	China	Singapore
Primary driver	Shipbuilding export	State smart-shipping plan	Port efficiency & safety
Flagship vessel	Prism Courage (LNGC)	Zhi Fei (300 TEU)	Autonomous tugs/harbour craft
Autonomy focus	Ocean Degree 2–3	Coastal Degree 3	Port Degree 2–3

USA, UK & Beyond: Industrial Policy, Defence, Rule-Making

🇺🇸 USA — MASGA & Navy USV

White House · MARAD · USCG · US Navy

- MASGA — "Make American Shipbuilding Great Again": 2025 Exec. Order & Maritime Action Plan; Korea pledges $150 bn (Hanwha Philly Shipyard, HD Hyundai–HII) — autonomy as a yard-modernisation lever
- Sea Hunter / Seahawk MDUSV (DARPA → Navy)
- Ghost Fleet Overlord: 4,400 nm autonomous transit
- Navy FY26: $2+ bn unmanned surface budget line
- SHIPS for America Act; USCG MASS policy & Saildrone/Sea Machines civil USVs

🇬🇧 UK — MAXCMAS & Maritime 2050

Lloyd's Register · QUB · Rolls-Royce · MCA

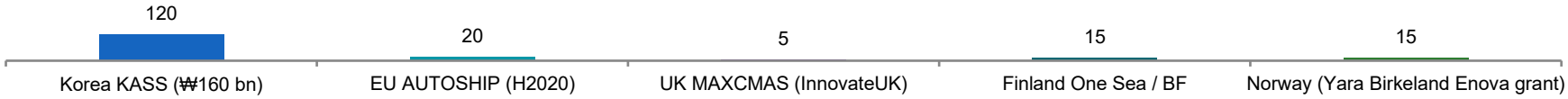
- MAXCMAS: Machine-Executable COLREGs for MASS — rule-based collision avoidance validated at sea
- Mayflower Autonomous Ship (IBM/ProMare) 2022 Atlantic crossing
- MCA MASS Code & Workboat Code Annex (2024)
- Plymouth Smart Sound 5G test range
- Lesson: COLREGs encoding is the certifiable core

🌍 Others worth watching

Australia · Israel · Turkey · Gulf

- Australia: Ocius Bluebottle solar/wind USVs
- Bulgaria: MTG Dolphin (located in Varna) developed a Medium Unmanned Surface Vehicle (MUSV) concept known as Proteus 36 MLC
- Türkiye: ULAQ & MARLIN armed USVs; MIR USV; ITU/YTU Autonomous Hydrographic Surveying Vessel (AHSV) research vessels
- UAE/Saudi: ADASI, NEOM autonomous port concepts
- Common thread: dual-use funding accelerates MASS

Reported public grants for named civil MASS programmes, ≈$M (open sources; JP/CN/US defence not comparable)



Who Builds Autonomy: The Industrial Landscape

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Full-stack autonomy & bridge systems

Avikus (HD Hyundai) KR

HiNAS Control / NeuBoat — 1st transoceanic autonomous LNGC voyage (Prism Courage, 2022); >200 ship orders

Samsung Heavy Ind. SAS KR

Samsung Autonomous Ship (SAS) — remote & autonomous navigation; sea-trialled on SHI-built LNG carriers and the Samsung test ship (2020–23)

Kongsberg Maritime NO

Autonomy stack for Yara Birkeland & ASKO; ex-Rolls-Royce Marine; K-Sim simulators

Wärtsilä FI

SmartMove / IntelliTug, Navi-Planner, remote pilotage (Folgefonn ferry auto-docking)

ABB Marine CHFI

Ability Marine Pilot Vision & Control — Suomenlinna II remote-piloted ferry (2018); Ro-Pax auto-docking

Sensors, navigation & ship-control electronics

Furuno JP

Radar/ECDIS, AIS, sensor-fusion & AI object recognition in MEGURI 2040 vessels

Nabtesco JP

Engine & propulsion remote-control, auto-berthing actuators in MEGURI 2040 (Soleil, Mikage)

JRC / Japan Radio JP

Collision-avoidance navigation displays, remote-monitoring links

Orca AI IL

Camera + AI watch-keeping assistant; deployed fleet-wide (NYK, MOL trials)

Sea Machines US

SM300/SM400 autonomy retrofit kits for tugs & workboats; AI-ris perception

Operators, USV makers & classification

Massterly (Kongsberg × Wilhelmsen) NO

1st autonomous-ship operating company; runs Yara Birkeland from Horten ROC

NYK · MOL · Mitsui E&S JP

MEGURI 2040 shipowner/yard partners; MOL "Mikage" container autonomy demo

Saildrone US · Ocius AU · Yunzhou CN

Wind/solar USVs — >1 M nm surveyed; Yunzhou 1,000+ units

Hanwha Ocean KR · CSSC CN

Shipyard in-house MASS programmes; Zhi Fei autonomous feeder

ClassNK · DNV · LR · BV · Türk Loydu

MASS guidelines, type approval, remote-ops notations — the gatekeepers of certification

Pattern: the winners pair a systems integrator (Avikus, Kongsberg, Wärtsilä) with sensor/actuator specialists (Furuno, Nabtesco) and a dedicated operator (Massterly) — under a class society's MASS notation.

The Autonomy Stack: Industry Delivery vs. R&D Frontier

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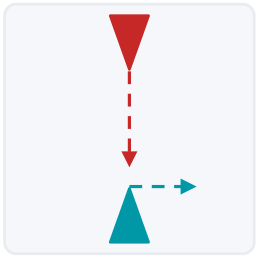
	1 Perception *radar · AIS · EO/IR · lidar · GNSS/IMU*	2 Situational awareness *tracking · CPA/TCPA · encounter classification*	3 Decision *COLREGs collision avoidance · path planning*	4 Control *autopilot · DP · engine/rudder*	5 Remote ops & assurance *ROC · cyber · class notation*
Industry today	▸ Furuno, JRC (radar/ECDIS) ▸ Orca AI (AI cameras) ▸ Sea Machines AI-ris	▸ Avikus HiNAS 2.0 ▸ Kongsberg Situational Awareness ▸ Wärtsilä Navi-Planner	▸ Avikus, Samsung SAS ▸ MAXCMAS rule-base (LR/RR) ▸ Yunzhou, Navigation Brilliance	▸ Nabtesco engine remote-control ▸ ABB, Kongsberg DP ▸ Wärtsilä SmartMove auto-docking	▸ Massterly ROC (Horten) ▸ MOL/NYK ROC Tokyo ▸ ClassNK, DNV, LR, BV, Türk Loydu
ITU R&D	▸ AHSV: RTK-GNSS 8 mm + IMU ▸ Simulator: synthetic AIS/radar traffic	▸ Encounter geometry → Rule 13/14/15 role ▸ Rule 7 risk from CPA + bearing drift	▸ MMG receding-horizon planner (~45 ms) ▸ Wind-feasible sailing supervisor	▸ LQR DP on AHSV ▸ PID/LQR track-keeping in simulator	▸ PIL/HIL V-model evidence ▸ AI in simulators for training & test
Open gap	Certified sensor-fusion performance in rain/clutter	Intent inference of human-crewed targets	Explainable, certifiable decisions; dynamics-feasibility	Degraded-mode fallback & fail-safe	Simulation-based certification acceptance

Harmonisation: industry has productised layers 1, 4 and 5; the R&D frontier — and ITU's contribution — is layer 3 plus the simulation-based assurance that lets class societies certify it.

COLREGs: The Legal Compass for Autonomous Ships

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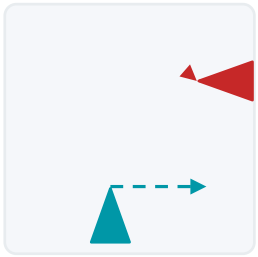
Rule 14 — Head-on



Both alter course to STARBOARD; neither is stand-on.

■ Own ship ■ Target

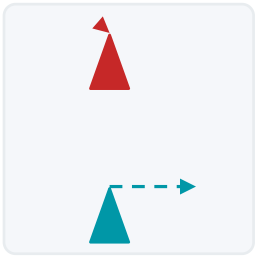
Rule 15 — Crossing



Own ship sees target on her STARBOARD bow → own ship gives way: alter to starboard, pass astern.

■ Own ship ■ Target

Rule 13 — Overtaking



Overtaking vessel (>22.5° abaft beam) keeps clear until past & clear.

■ Own ship ■ Target

Rule	Requirement	What it means for the autonomy stack
5 Look-out	Proper look-out by sight, hearing & all available means	Sensor fusion (radar/AIS/EO-IR/lidar) must match or exceed human perception → certifiable perception
6 Safe speed	Speed adapted to visibility, traffic, manoeuvrability	Speed governor as a hard constraint in the planner — our MMG model gives true stopping/turning ability
7 Risk of collision	Use radar plotting; assume risk if compass bearing is steady	CPA/TCPA computation on every target; ambiguity ⇒ assume risk
8 Action to avoid	Early, substantial, readily apparent; avoid small successive changes	Cost-function penalty on small heading changes → "obvious" manoeuvres (our w_4 term)
17 Stand-on	Keep course & speed, but may act when give-way fails	Supervisor layer: timed escalation from stand-on to evasive mode

UK MAXCMAS & Japan Meguri both proved: machine-executable COLREGs is the certifiable core — our planner embeds Rules 6, 8, 13–17 as cost terms.

Black Sea Maritime Autonomy Ecosystem — Where We Stand

TR Türkiye — Istanbul

- ITU MBD&C Lab: MMG planner, sailing supervisor, PIL
- ITU–YTU AHSV research vessel
- Bosphorus simulator & TÜBİTAK AI-sim project
- Industry: ULAQ/MARLIN USVs, Türk Loydu MASS rules

TR Türkiye — Policy

- Montreux & Bosphorus VTS expertise
- KEGM/Türk Loydu MASS guidelines (draft)
- Shipyard base: Tuzla & Yalova

Gap

No shared autonomous-ship test area or MASS sandbox in the Black Sea — unlike Jaakonmeri (FI), Horten (NO), Ulsan (KR).

Opportunity

Enclosed sea, dense Bosphorus traffic, Danube corridor and EU neighbourhood funds = ideal regulated test-bed.

BLACK SEA

≈ 436,000 km² · 6 coastal states · Bosphorus: 40,000+ transits/yr

BG Bulgaria — Varna

- CONNEXUS forum & Marine Cluster
- Nikola Vaptsarov Naval Academy simulators
- Black Sea Maritime Cluster
- EU funding access (Horizon, Interreg NEXT)

RO Romania — Constanța

- Largest Black Sea port; Danube link
- CERONAV training & simulator centre
- EU Danube inland-autonomy pilots

GE Georgia

- Middle Corridor (TITR) logistics
- Poti / Anaklia port modernisation

UA RU Northern littoral

- Conflict accelerated USV tech
- Post-war demining & survey need

Proposal: a Black Sea MASS Test Corridor (Varna–Constanța–Istanbul) with shared simulator scenarios, a joint COLREGs rule-base and open AHSV-class survey drones.

Key Takeaways & Thank You

1. **Autonomy is global & accelerating**

   Avikus, Samsung SAS, Kongsberg/Massterly, Furuno & Nabtesco have moved from pilots to commercial Degree 2–3 service; the IMO MASS Code sets the clock.

2. **IMO Roadmap**

   MASS Code, COLREGs & Readiness of Class Companies, Industry, Institutes, Rule Making Organizations

3. **Key Technologies for Autonomous Ships**

   Sensor Fusion, Mathemtical Modeling, V-Cycle Approach,

4. **COLREGS Collision Avoidance Regulations**

   COLREGs: The Legal Compass for Autonomous Ships, Rule 14 — Head-on, Rule 14 — Crossing, Rule 13 — Overtaking

5. **The Black Sea can lead regionally**

   Simulator + AHSV + CONNEXUS network → a Varna–Constanța–Istanbul MASS test corridor.

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[1] Jambak & Bayezit, Ocean Engineering, 2026.
[2] Jambak, Bayezit & Reyhanoglu, Drones, 2026.
[3] Jambak, Abdelkhalek, Bayezit, Kinaci — Ship Simulator for Control Testing.
[4] Kinaci — Digitalization Framework in Ship Navigation Autonomy, WMU.


“Never be afraid to ask questions,,
because if you never ask, the answer is always no.“
Sheena Yap Chan
THANK YOU
ISMAIL BAYEZIT
ISTANBUL TECHNICAL UNIVERSITY